Amazon Web Services: Amazon Elastic Compute Cloud (EC2)

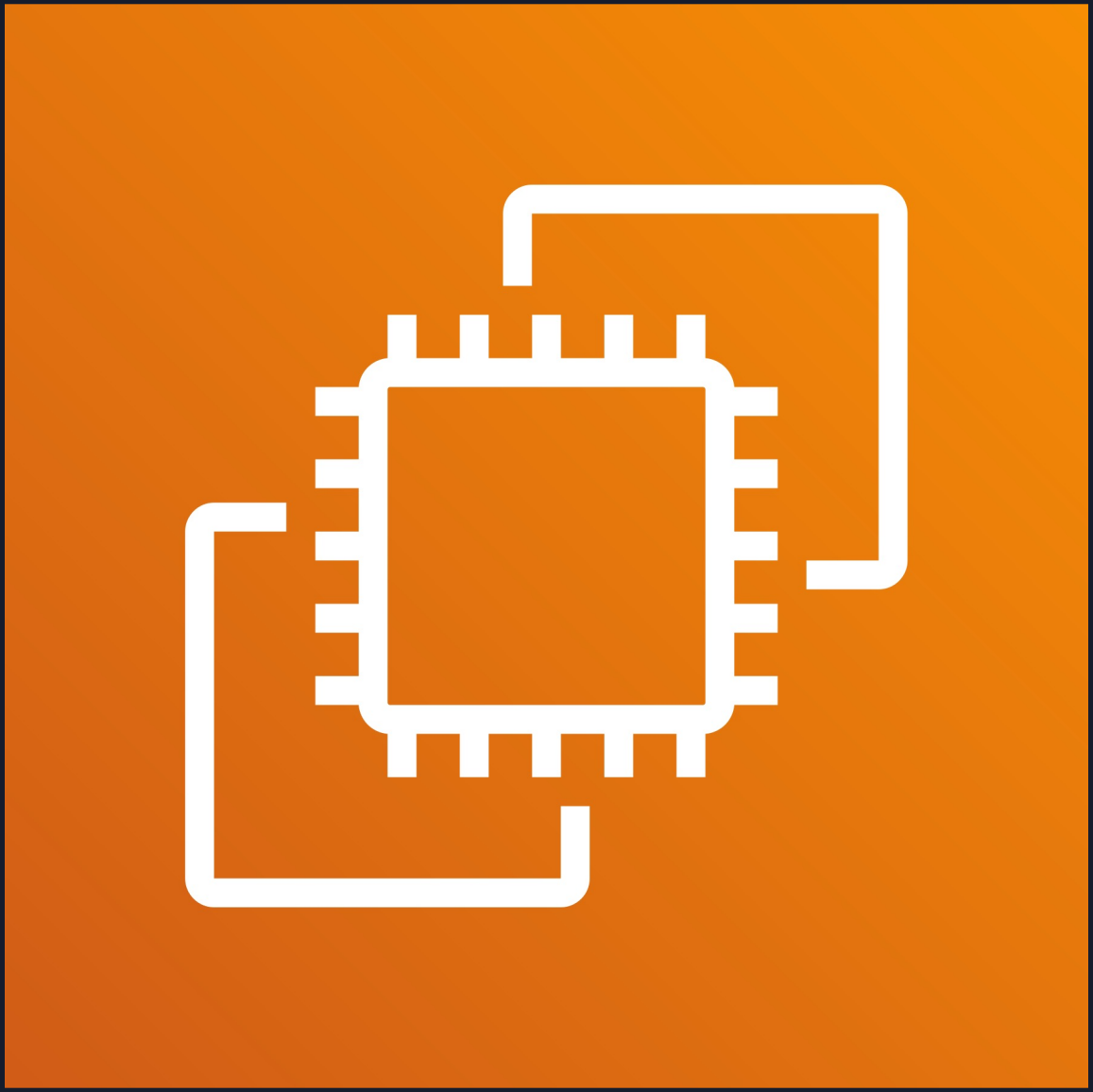
Topics

- What is EC2?
- Pricing Options
- Using Roles
- Security Groups and Bootstrap scripts
- EC2 Metadata and User Data
- Networking with EC2
- AWS Outposts

What is EC2?

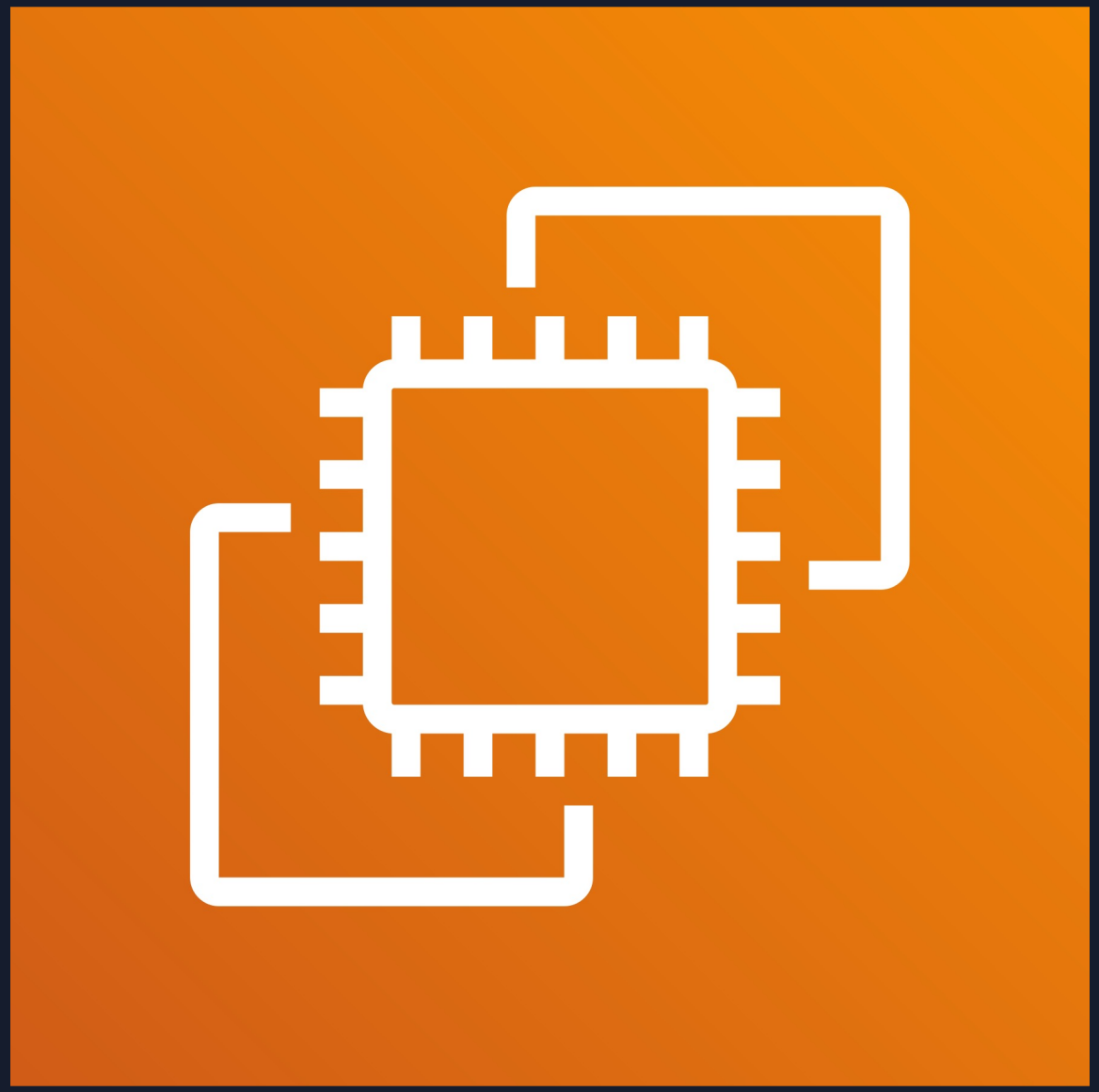
What is EC2?

- Stands for Elastic Compute Cloud
- A virtual machine or server that is hosted in AWS instead of a company's own data center
- Gives you the capacity that you want when you need it
- You have complete control of your instances



What is EC2?

- Launches in 2006
- Became a game changer
- Pay only for what you use
- Don’t waste capacity
- Your EC2 instances are up in minutes



Pricing Options

How is EC2 priced?

"Although the number and types of services offered by AWS have increased dramatically, our philosophy on pricing has not changed. You pay as you go, pay for what you use, pay less as you use more, and pay even less when you reserve capacity. All these options are empowering AWS customers to choose their preferred pricing model and increase flexibility of their cost strategy."

https://docs.aws.amazon.com/whitepapers/latest/how-aws-pricing-works/how-aws-pricing-works.pdf

4 EC2 Pricing Models

- **On-Demand Instances** – pay a fixed rate by the hour (or second) with no long-term commitment
- **Savings Plans** – capacity reservation with significant discount of the hourly charge for an instance; pay upfront; contract terms are 1 year to 3 years
- **Spot** – bid the price that you want for unused instance capacity; greater savings if applications have flexible start and end times
- **Reserved Instances (RI)** – significant discount compared on On-Demand Instance pricing; assigned to specific AZ; confident that you will get the capacity you need when you need it (**On-Demand Capacity Reservations**)
- **Dedicated hosts** – physical EC2 servers dedicated for your use; reduce cost by allowing users to use existing server-bound SW licenses

https://aws.amazon.com/ec2/pricing/

On-Demand Instances Use Cases

- Users who prefer the low cost and flexibility of Amazon EC2 without upfront payment or long-term commitments
- Applications with short-term, spiky, or unpredictable workloads that cannot be interrupted
- Applications being developed or tested on Amazon EC2 for the first time

Savings Plans Use Cases

- Workloads with consistent and steady-state usage
- Customers who want to use different instance types and compute solutions across different locations
- Customers who can make a monetary commitment to use compute services over a one or three-year term

Spot Instances Use Cases

- Applications that have flexible start and end times
- Applications that are only feasible at very low compute prices
- Users with fault-tolerant and/or stateless workloads
- Are not good for persistent workloads, critical jobs, databases

Reserved Instances Use Cases

- Business-critical events or workloads that require capacity assurance
- Workloads that need to meet regulatory requirements for high availability
- Disaster recovery

Dedicated Hosts Use Cases

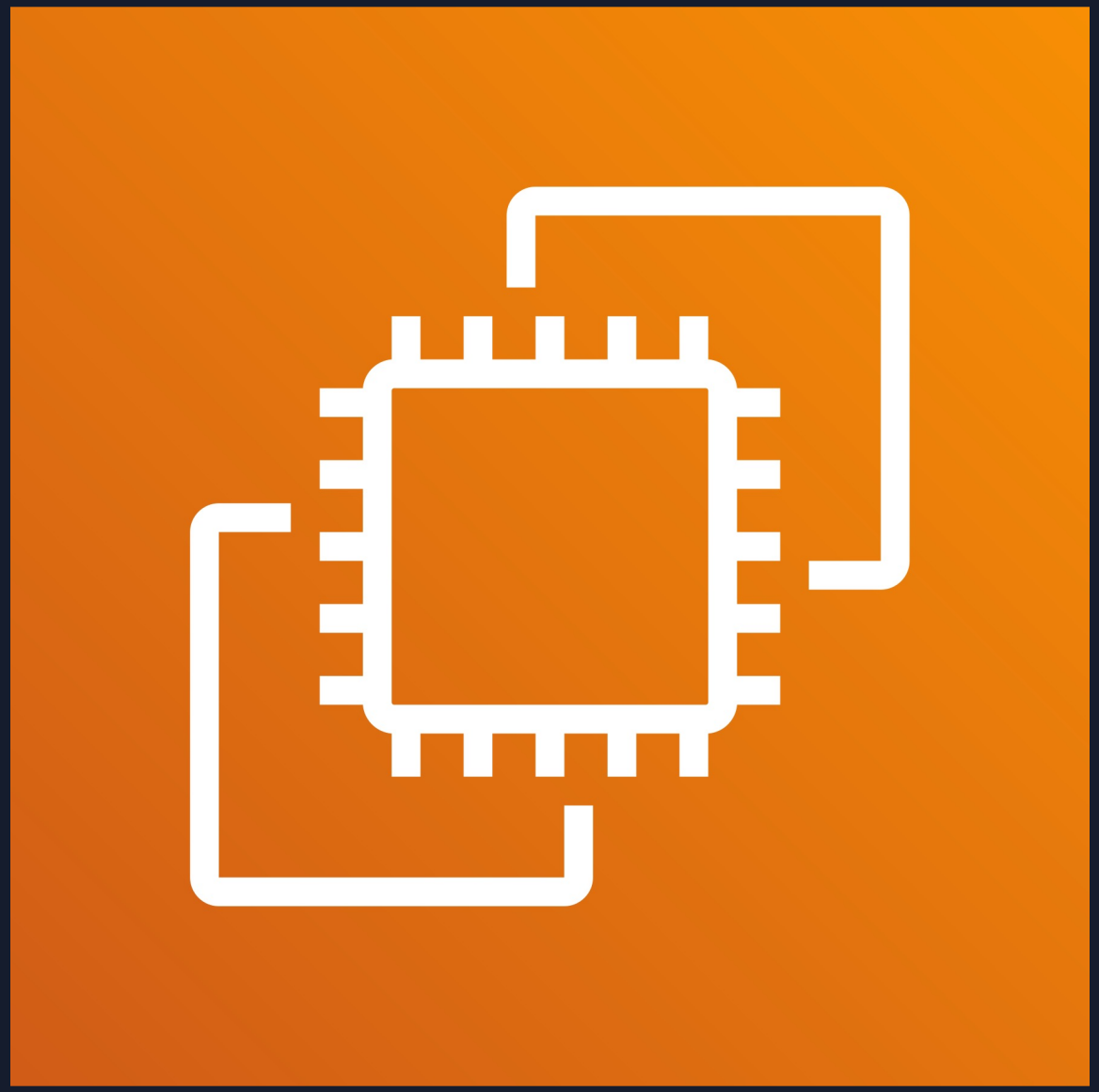
- Regulatory requirements that may not support multi-tenant virtualization
- Licensing that does not support multi-tenancy or cloud deployments

EC2 Instance Types

Instance Family	Description	Instance Type
General Purpose	• Provide a balance of compute, memory, and networking resources • Support diverse workload • Ideal for web servers and code repos	M7a, M7g, Mac, T4g, T3, T3a, T2, M6g, M6i, M6a, M5, M5a, M5n, M5zn, M4, A1
Compute Optimized	• Ideal for compute bound applications that benefit from high performance processors • Suited for batch processing workloads, media transcoding, high performance computing, scientific modeling, dedicated gaming servers, machine learning inference and other compute intensive applications	C7g, C6g, C6gn, C6i, C6a, Hpc6a, C5, C5a, C5n, C4
Memory Optimized	• Designed to deliver fast performance for workloads that process large data sets in memory	R6g, R6i, R5, R5a, R5b, R5n, R4, X2gd, X2idn, X2iedn, X2iezn, X1e, X1, High Memory, z1d
Accelerated Computing	• Use hardware accelerators, or co-processors, to perform functions, such as floating-point number calculations, graphics processing, or data pattern matching, more efficiently than is possible in software running on CPUs.	P4, P3, P2, DL1, Trn1, Inf1, G5, G5g, G4dn, G4ad, G3, F1, VT1
Storage Optimized	• Designed for workloads that require high, sequential read and write access to very large data sets on local storage • Optimized to deliver tens of thousands of low-latency, random I/O operations per second (IOPS) to applications.	Im4gn, Is4gen, I4i, I3, I3en, D2, D3, D3en, H1

https://aws.amazon.com/ec2/instance-types/

Demo: Launch an EC2 Instance AWS CLI

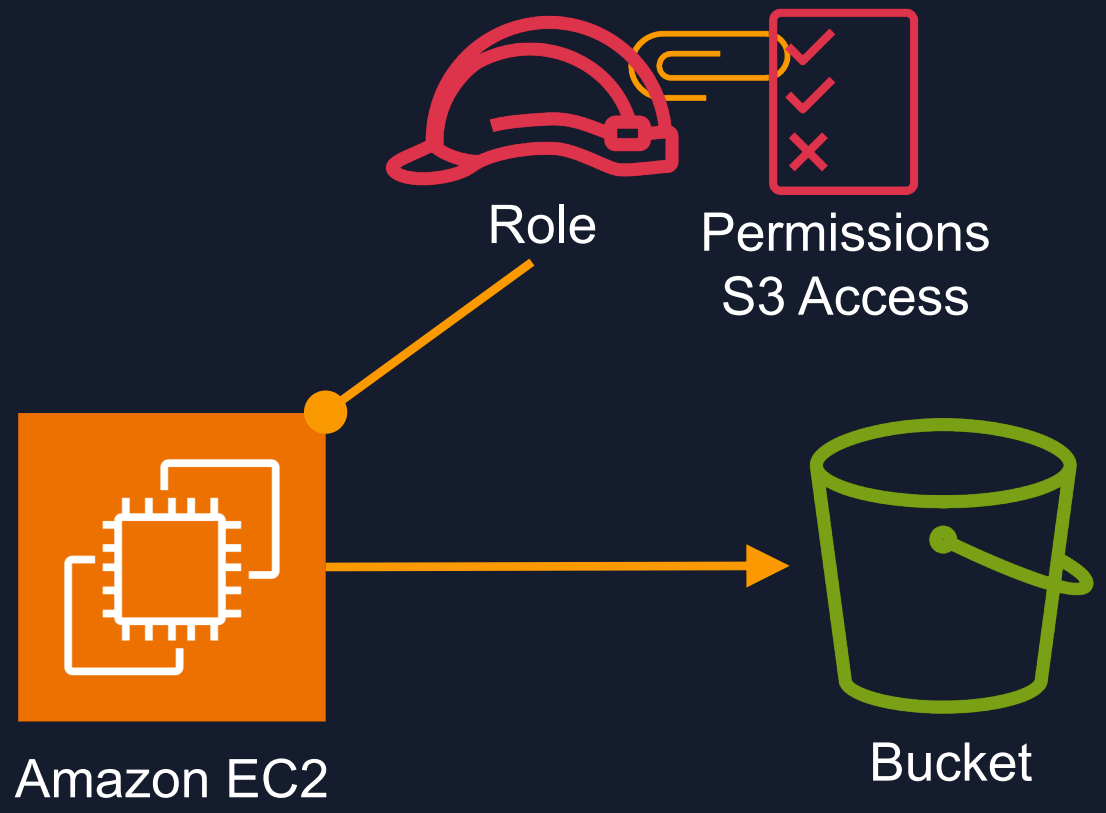


Using Roles

What is an IAM Role?

- An identity that you can create in IAM that has specific permissions
- Similar to a user, it is an AWS identity with permission policies that determine what the identity can and cannot do
- Instead of being associated with one person, a role is intended to be assumable by anyone who needs it
- Are temporary
- Can be assumed by people, AWS architecture or other system level accounts
- Preferred option from a security perspective

Demo: IAM Roles



Security Groups and Boostrap Scripts

How Computers Comminucation

Linux
SSH
Port 22

Windows
RDP
Port 3389

HTTP
WEB
BROWSING
Port 80

HTTPS
ENCRYPTED
WEB
BROWSING
(SSL)
Port 443

Security Groups

- Virtual firewalls for you EC2 instances
- Can place any number of EC2 instances with a security group
- Can have multiple security groups attached to an EC2 instance
- By default, all inbound traffic is blocked, and all outbound traffic is allowed
- To let everything in use IP address range 0.0.0.0/0
- In production, you'd only open 0.0.0.0/0 to port 80 (HTTP) and to port 443 (HTTPS)

Bootstrap Scripts

```
#!/bin/bash
# install apache
yum install httpd -y
# start apache
yum service httpd start
```

Demo: Security Groups and Bootstrap Scripts

EC2 Metadata and User Data

What is EC2 Metadata?

- EC2 metadata is data about your EC2 instance
- Includes information such as private IP addresses, public IP addresses, the hostname of your EC2 instance, what security groups it's in and so on
- To retrieve metadata, use the curl command

```
TOKEN=`curl -X PUT "http://169.254.169.254/latest/api/token" -H "X-aws-ec2-metadata-token-ttl-seconds: 21600"` \
&& curl -H "X-aws-ec2-metadata-token: $TOKEN" -v http://169.254.169.254/latest/meta-data/
```

- To retrieve user data, replace meta-data with user-data

From: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/instancedata-data-retrieval.html

Demo: EC2 Metadata

```
#!/bin/bash
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd
cd /var/www/html
echo "<html><body><h1>My IP is" > index.html
TOKEN=$(curl -s -X PUT "http://169.254.169.254/latest/api/token" -H
"X-aws-ec2-metadata-token-ttl-seconds: 21600")
PUBLIC_IP=$(curl -s -H "X-aws-ec2-metadata-token: $TOKEN"
http://169.254.169.254/latest/meta-data/public-ipv4)
echo "$PUBLIC_IP" >> index.html
echo "</h1></body></html>" >> index.html
```

Networking with EC2

Virtual Networking Options

ENI
Elastic Network Interface

For basic day-to-day networking

EN
Enhanced Networking

Uses single root I/O virtualization to provide high performance

EFA
Elastic Fabric Adapter

Accelerate HPC and machine learning applications

ENI – Elastic Network Interface

ENI

Elastic Network Interface

For basic day-to-day networking

- Is a virtual network card that allows the following options:
  - Private IPv4 address
  - Public IPv4 address
  - MAC Address
  - 1 or more security groups
  - Many IPv6 addresses
- Use cases:
  - Create a management network
  - Use network and security appliances in VPC
  - Low budget, high availability solution
- An EC2 instance will have an ENI attached to it by default

EN – Enhanced Networking

EN

Enhanced Networking

Uses single root I/O virtualization to provide high performance

- For high performance networking between 10 gigabits per second and 100 gigabits per second
- Uses single root I/O virtualization or SR-IOV to provide higher I/O performance and lower CPU utilization
- Provides higher bandwidth, higher packets per second performance, and it gives you a consistently lower inter-instance latency.

EFA – Elastic Fabric Adapter

EFA

Elastic Fabric Adapter

Accelerate HPC and machine learning applications

- Network device that you can attach to your EC2 instances to accelerate HPC and machine learning applications
- Provides lower and more consistent latency and higher throughput than TCP transport

AWS Outposts

AWS Outposts family

What is Outposts?

- Brings the AWS data center on-premises (server racks)
- Allows you to have a hybrid cloud where you can use AWS services inside your data center
- It is a fully managed infrastructure by AWS
- Brings the AWS Management Console, APIs, and SDKs into your data center
- Two Outposts family members:
  - Outposts Rack
  - Outposts Servers

AWS Outposts rack

AWS Outposts servers

Checkpoint

Which statement best defines an EC2 instance?

1. It is a service that provides simple and easy to use messaging queues.
2. It is a service that provides secure, resizable compute capacity in the cloud.
3. It is a service that allows you to bridge the gap securely and easily between your physical data center and the AWS Cloud.
4. It is a service that provides durable and reliable remote object-based storage.

What is user data commonly used for?

1. For updating the content of your S3 bucket
2. For bootstrapping an EC2 instance as it comes online
3. For deploying EC2 instances into an Auto Scaling group
4. For rotating your IAM passwords

Which of the following are valid use cases for IAM roles?

Choose 3

1. Attaching a role to an EC2 instance to allow it to talk to other AWS resources.
2. Providing access to your AWS resources to users that are authenticated using an external identity provider.
3. Allowing a user from another AWS account to assume a role in your account and interact with your AWS architecture.
4. Defining a set of permissions for an IAM role, and directly associating that role with an IAM group.

Why would you want to spin up an EC2 Dedicated Host?

1. Because you need to create a cluster of EC2 instances
2. Because you have sensitive workloads that you don't want AWS to see
3. Because your boss told you to
4. Because your application has hardware-specific licensing requirements

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